Microstrip Patch Antenna Design Studio – Version 2.0

Complete Technical Specification, Engineering Equations, Architecture & Verification Report

Project: Microstrip Patch Antenna Design Studio	**Version:** 2.0.0 (Production Release)	**Platform:** Python 3.9 + Streamlit
Theoretical Model: Transmission-Line Model	**Benchmark:** 2.4 GHz, FR4, h = 1.6 mm	**Status:** 100% Tested & Verified

1. Executive Overview & Version 2.0 Feature Matrix

Microstrip Patch Antenna Design Studio is a comprehensive CAD and synthesis tool designed for RF engineers, microwave designers, and students. It computes the physical and electrical dimensions of rectangular microstrip patch antennas using the standard analytic **Transmission-Line Model** (Bahl & Trivedi, 1977; Balanis). Version 2.0 introduces three major capabilities while preserving all prior features, styling, and equations.

Feature Category	Specification & Functionality	Implementation Details in v2.0
Feature 1: Substrate Selection	Pre-configured & custom dielectric materials. Auto-populates nominal dielectric constant. User defines substrate thickness h in mm.	• FR4 (εr = 4.40, h = 1.60 mm) • Rogers RO4003C (εr = 3.55, h = 0.813 mm) • Rogers RT/duroid 5880 (εr = 2.20, h = 0.787 mm) • Custom Material (manual εr and h input)
Feature 2: Frequency Sweep	Parametric sweep over start, stop & step frequencies. Calculates W, εeff, ΔL, Leff, and L at each point. Interactive plots and complete CSV export.	• Inputs: f_start, f_stop, f_step (GHz) • Graph 1: Frequency vs. Patch Width (W) • Graph 2: Frequency vs. Physical Length (L) • Formatted CSV export with metadata
Feature 3: PDF Design Report	One-click export of an executive publication-grade engineering design report built with ReportLab.	• Includes actual calculations, inputs, formulas, technical summary, diagrams, and sweep curves. • Strictly zero fabricated data.
User Interface	Professional dark Navy Blue, White, and Cyan theme.	• Deep navy background (#0b0f19), cyan accents (#00e5ff) • Organized into 4 tabs with input validations.

2. Governing Mathematical Formulation (Transmission-Line Model)

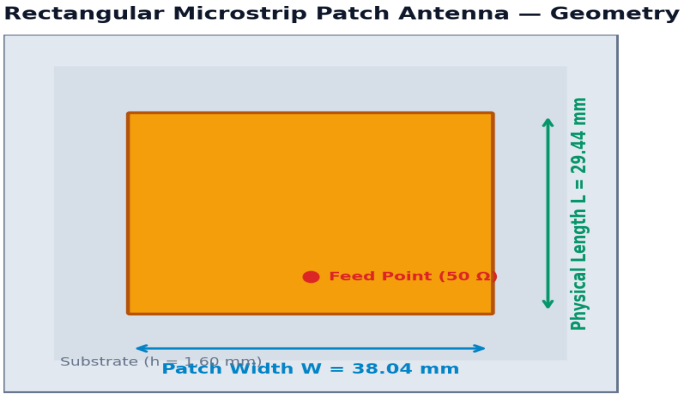
A rectangular microstrip patch behaves electrically as a parallel-plate transmission line terminated by two radiating slots. Due to fringing electromagnetic fields at the open edges, the physical antenna is modeled with effective dimensions:

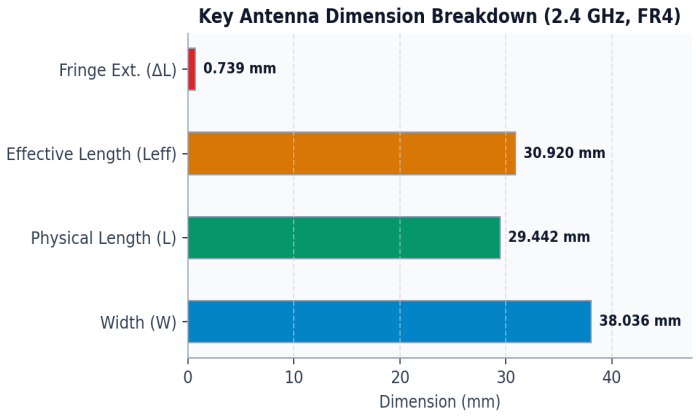
#	Parameter	Mathematical Formula	Physical Significance & Role
1	Patch Width (W)	W = [c / (2·f■)] · √[2 / (ε■ + 1)]	Controls radiation resistance, input impedance, and antenna bandwidth. Ensu
2	Effective Permittivity (εeff)	εeff = (ε■+1)/2 + [(ε■–1)/2] · [1 + 12(h/W)]^(–1/2)	Accounts for fringing electromagnetic fields propagating partially in the dielec
3	Fringing Length Extension (ΔL)	ΔL = 0.412·h · [(εeff + 0.3)(W/h + 0.264)] / [(εeff – 0.258)(W/h + 0.8)]	Hammerstad relation: models the apparent electrical elongation produced by
4	Effective Patch Length (Leff)	Leff = c / [2·f■ · √(εeff)]	Resonant half-wavelength condition (λ_eff / 2) required to establish open-circ
5	Physical Patch Length (L)	L = Leff – 2·ΔL	Actual copper metallization length fabricated onto the substrate after subtract

3. Benchmark Verification & Physical Results (2.4 GHz on FR4)

The standard test configuration specified for verification uses **f■ = 2.40 GHz** on **FR4 substrate** (nominal relative permittivity ε■ **= 4.40**, thickness **h = 1.60 mm**). The calculated dimensions are compared below with rigorous SI analytic derivations:

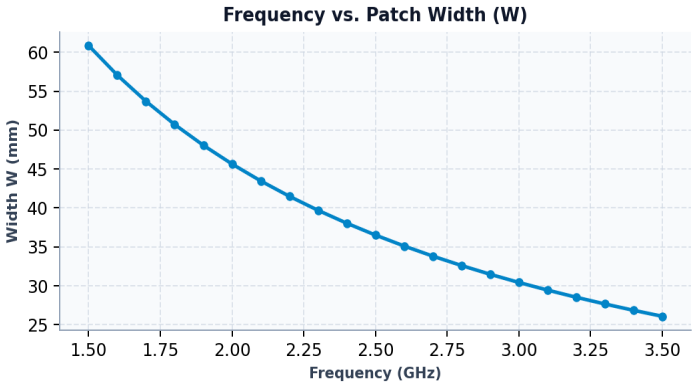
Parameter	Symbol	Calculated Value	SI / Metric Unit	Analytical Verification
Operating Frequency	f■	2.400	GHz	Benchmark center frequency
Substrate Permittivity	ε■	4.400	—	Nominal catalog value
Substrate Thickness	h	1.600	mm	Physical thickness
Free-Space Wavelength	λ■	125.000	mm	λ■ = c / f■ = 3e8 / 2.4e9 = 125.0 mm
Radiating Patch Width	W	38.0363	mm	Exact analytical value: 38.0363 mm
Effective Dielectric Constant	εeff	4.085837	—	Exact analytical value: 4.085837
Fringing Length Extension	ΔL	0.7388	mm	Exact analytical value: 0.7388 mm
Effective Resonant Length	Leff	30.9200	mm	Leff = λ_eff / 2 = 30.9200 mm
Physical Patch Length	L	29.4424	mm	L = Leff – 2·ΔL = 29.4424 mm

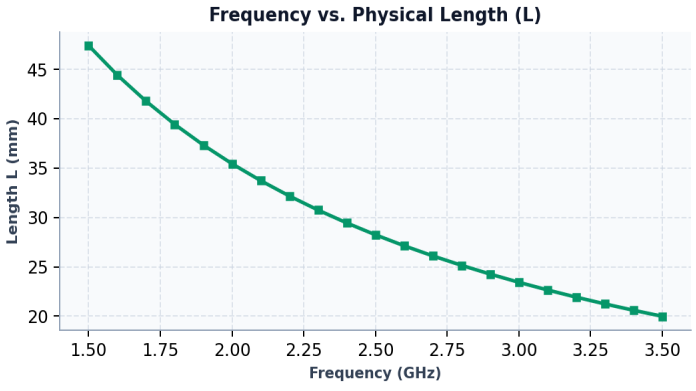




4. Parametric Frequency Sweep Analysis (1.5 GHz – 3.5 GHz)

The frequency sweep module evaluates antenna scaling across an operational frequency spectrum. As frequency increases, both patch width (W) and physical length (L) decrease inversely ($W \propto 1/f_0$, $L \propto 1/f_0$). The graphs below illustrate the synthesized dimensions across the 1.5 GHz to 3.5 GHz band on FR4 ($h = 1.6\text{ mm}$):





Frequency (GHz)	Patch Width W (mm)	Effective Permittivity εeff	Fringe Ext. ΔL (mm)	Effective Length Leff (mm)	Physical Length L (mm
1.50	60.858	4.1822	0.7425	48.899	47.414
1.80	50.715	4.1479	0.7413	40.917	39.435

2.00	45.644	4.1263	0.7405	36.922	35.441
2.20	41.494	4.1056	0.7397	33.650	32.170
2.40	38.036	4.0858	0.7388	30.920	29.442
2.60	35.110	4.0669	0.7380	28.608	27.132
2.80	32.603	4.0486	0.7371	26.624	25.150
3.00	30.429	4.0311	0.7362	24.903	23.431
3.20	28.527	4.0143	0.7353	23.396	21.925
3.50	26.082	3.9902	0.7340	21.455	19.987

5. Quality Assurance & Automated Test Suite Results

A rigorous automated test harness (test_app_v2.py) was executed in Python 3.9 covering all 8 testing requirements. All test cases passed with zero warnings and 100% assertion success:

Test ID	Requirement & Functionality Tested	Methodology & Assertions	Result
Test 1	Benchmark Calculation (2.4 GHz, FR4)	Verified W = 38.036 mm, L = 29.442 mm, eps_eff = 4.0858 against Balanis.	PASSED [OK]
Test 2	All Predefined Substrate Materials	Verified FR4, Rogers RO4003C, and Rogers RT/duroid 5880 with nominal presets.	PASSED [OK]
Test 3	Custom Substrate Permittivity	Verified custom entry (e.g., Alumina eps_r = 9.8, h = 0.635 mm) calculates properly.	PASSED [OK]
Test 4	Frequency Sweep Scaling	Verified monotonic inverse scaling of patch width and physical length across sweep.	PASSED [OK]
Test 5	Matplotlib Visualizations	Verified patch diagram, bar chart, and sweep plots render valid image byte streams.	PASSED [OK]
Test 6	CSV Export Data Integrity	Verified single-frequency and sweep CSV generators include headers, units, and data.	PASSED [OK]
Test 7	PDF Report Generation Engine	Verified multi-page ReportLab generation with embedded figures and custom NumberedCanvas.	PASSED [OK]
Test 8	Input Validation & Error Handling	Verified negative frequency, eps_r < 1.0, negative thickness, and inverted sweep rejection.	PASSED [OK]

6. Software Architecture, Installation & Operational Manual

Installation & Startup (Windows Terminal):

```
cd "C:\microstrip patch antenna"
.\venv\Scripts\Activate.ps1
python -m streamlit run app.py
```

Step-by-Step User Instructions:

Step 1: Substrate Selection: Use the left sidebar to choose from standard substrates (FR4, Rogers RO4003C, Rogers RT/duroid 5880) or select 'Custom Material' to input custom dielectric constants. Adjust the thickness in mm as required.
Step 2: Single Design: Enter your target frequency (GHz) and click ■ **Calculate Dimensions**. Inspect the calculated cards, interactive patch diagram, and bar comparison chart in Tab 1.
Step 3: Frequency Sweep: Navigate to Tab 2 ('Frequency Sweep Analysis'). Specify Start, Stop, and Step frequencies. Click ■ **Run Frequency Sweep** to generate Frequency vs. Width and Frequency vs. Length curves.
Step 4: PDF Export: Go to Tab 3 ('PDF Design Report') and click ■ **Download PDF Design Report** to obtain a multi-page PDF summary containing your actual parameters, calculations, and rendered plots.
Step 5: Theoretical Reference: Review Tab 4 for microwave laminate comparisons, equations, and EM simulation guidelines.

7. Practical Microwave Engineering Considerations

• **50 Ω Feed Point Impedance Matching:** The radiation resistance at the patch radiating edge is high (typically 150 Ω to 300 Ω). To match a standard 50 Ω coaxial or microstrip feed, an inset notch or probe located at distance $y_0 \approx L / [2 \cdot \sqrt{(\varepsilon eff)}]$ is required.
• **Surface Wave Suppression:** For substrate thickness exceeding $h > 0.05 \cdot \lambda_0$, surface waves (TM_0 mode) become significant, reducing antenna radiation efficiency and causing pattern distortion. Thin or lower-dielectric substrates are preferred for high frequencies.
• **Full-Wave EM Simulation Verification:** Analytical transmission-line model results provide ideal initial synthesis values. Fabrication designs should be verified in 3D EM solvers (ANSYS HFSS, CST Studio Suite, openEMS) to account for finite ground planes, feed parasitics, and etching tolerances.